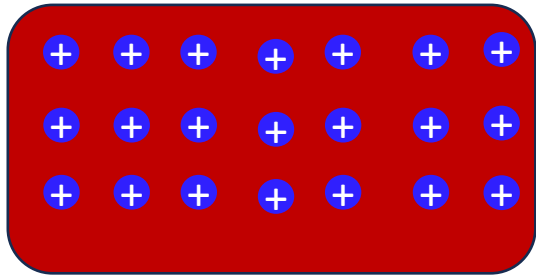
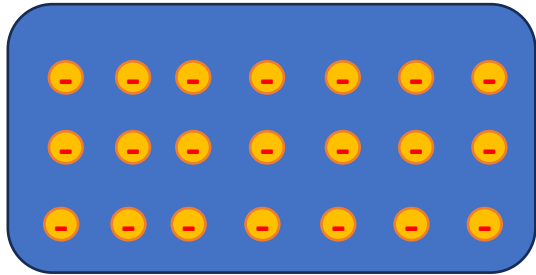


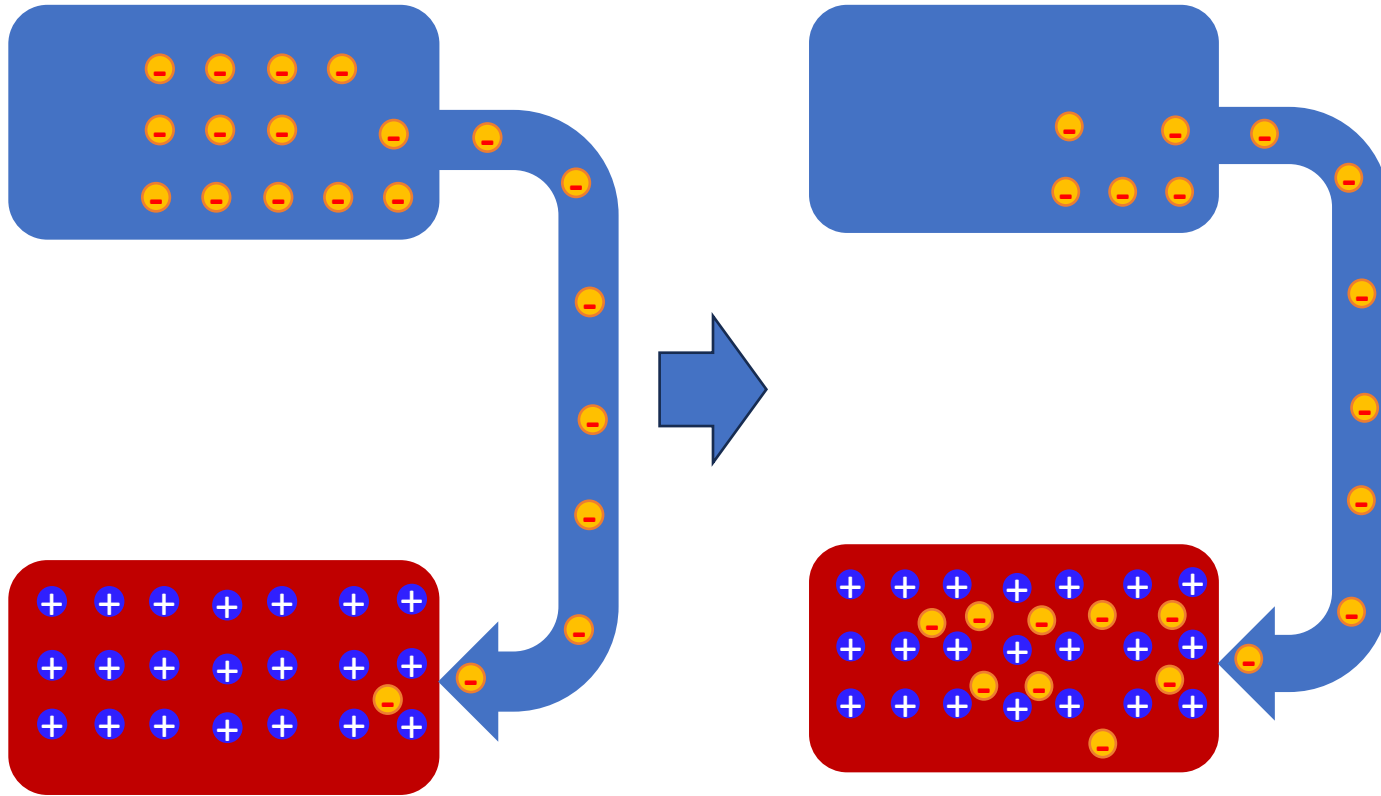
The story of potential difference and the flow of charge.



Situation one..

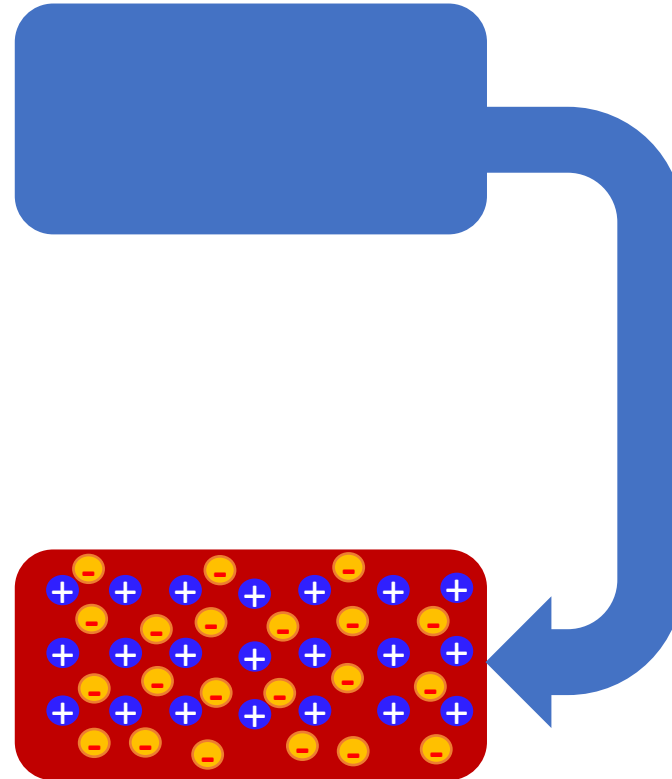
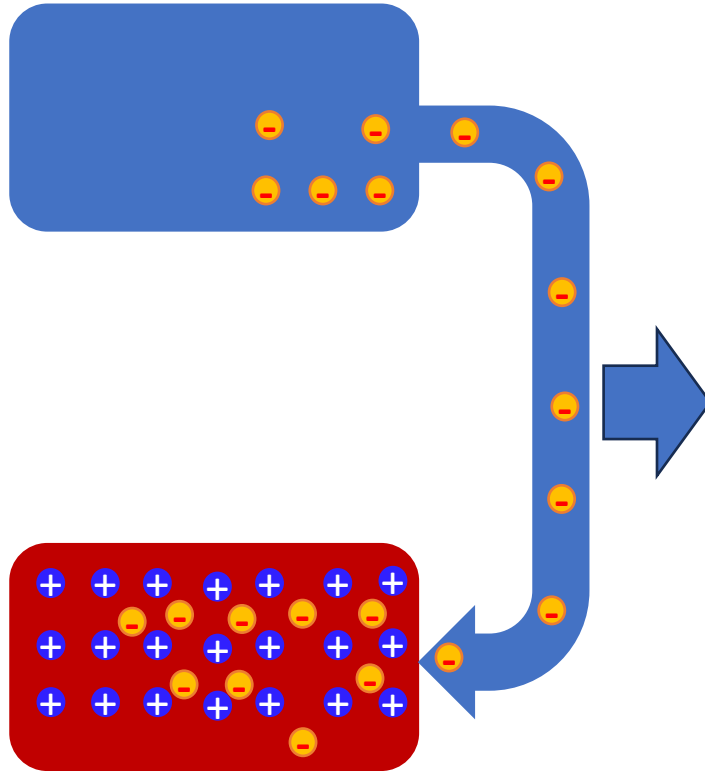
- Imagine there are two piles of charges contained in two locations, negative and positive, separated from each other. Lets consider that the negative charges are free to move within the container while the positive charges are fixed in their locations.
- Negative charges feels the pressure to run towards the positive charges. But they can't as the charges can't escape the container.
- So we can say there always exists a pressure on the electrons to move from negative container to the positive one. This is known as the potential difference between the containers also electromotive forces (EMF)
- What is the amount of charge transfer? Zero!! (no electrons can run to the positive piles despite there exists an emf.

Situation two..



- Let's connect the two piles with a piece of conductor.
- As the two piles are connected through the conductor, the electrons in the negative piles have found a door to release the pressure by running towards the positive charge piles.
- Or we can say the emf established between the two piles pushes electrons to the positive piles.
- So we can say that there is a flow of charges from negative container to the positive one.

Situation two:continues..

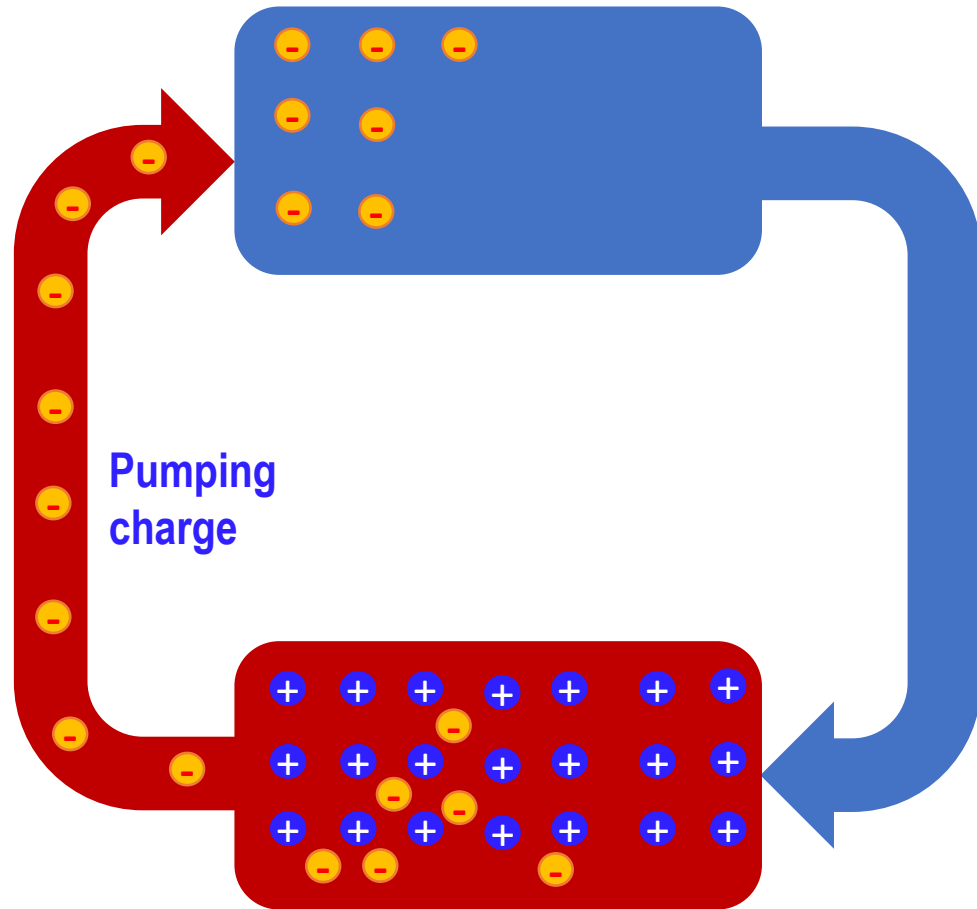


- This leads to the decrease in the number of electrons in the negative container and addition of electrons in the positive container.
- Eventually, there won't be any electrons left in the negative container as all of them moved to the positive container.

So what is the status of total charges in the two container?

- In negative container, charge is Zero
- and in the positive container total is Zero as well. Why? Because negative charges have neutralised the positive charges.

What is the potential difference or the emf between the two containers now?
None!! No separation of charges exist.



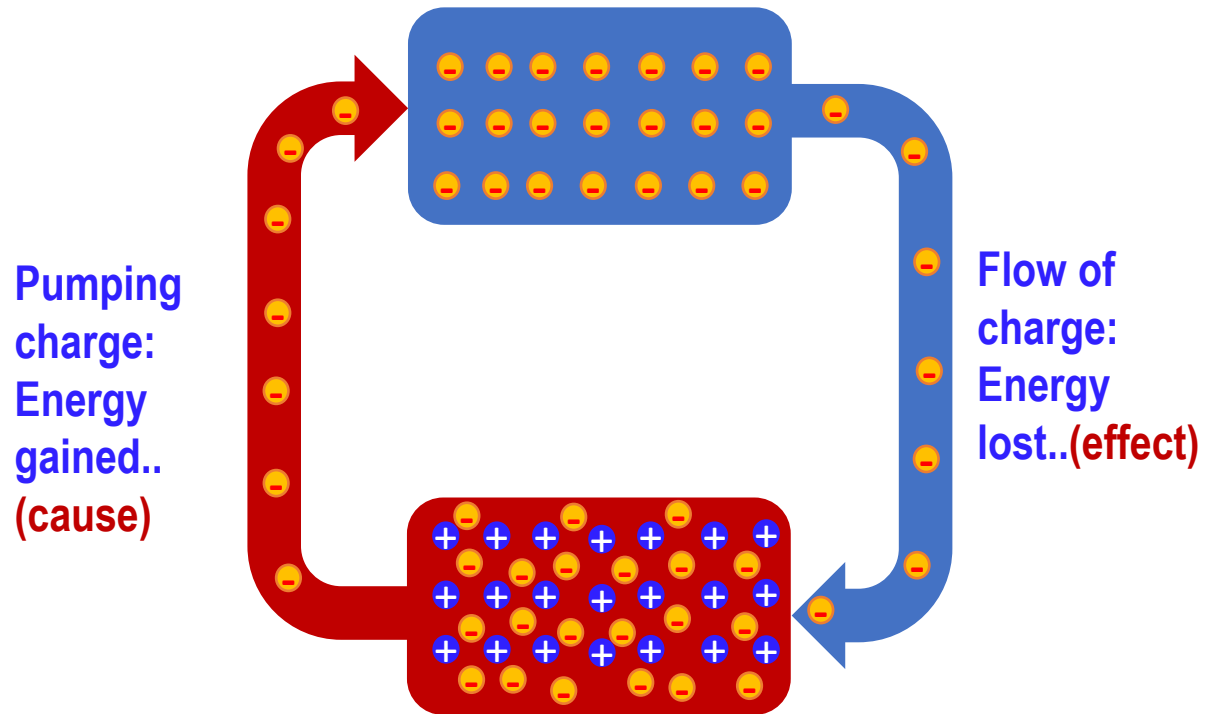
Situation three...

What do I have to do if I want to maintain the flow of charges from the negative pile to the positive?

Ans:

- Remove the negative charges from the lower container and populate the upper container.
- However, once the charges (negative and positive) are together, they don't like to be separated at all. So, what is the way?
- You have to do some work to separate them and pump the charges back up the upper container to establish the emf again.

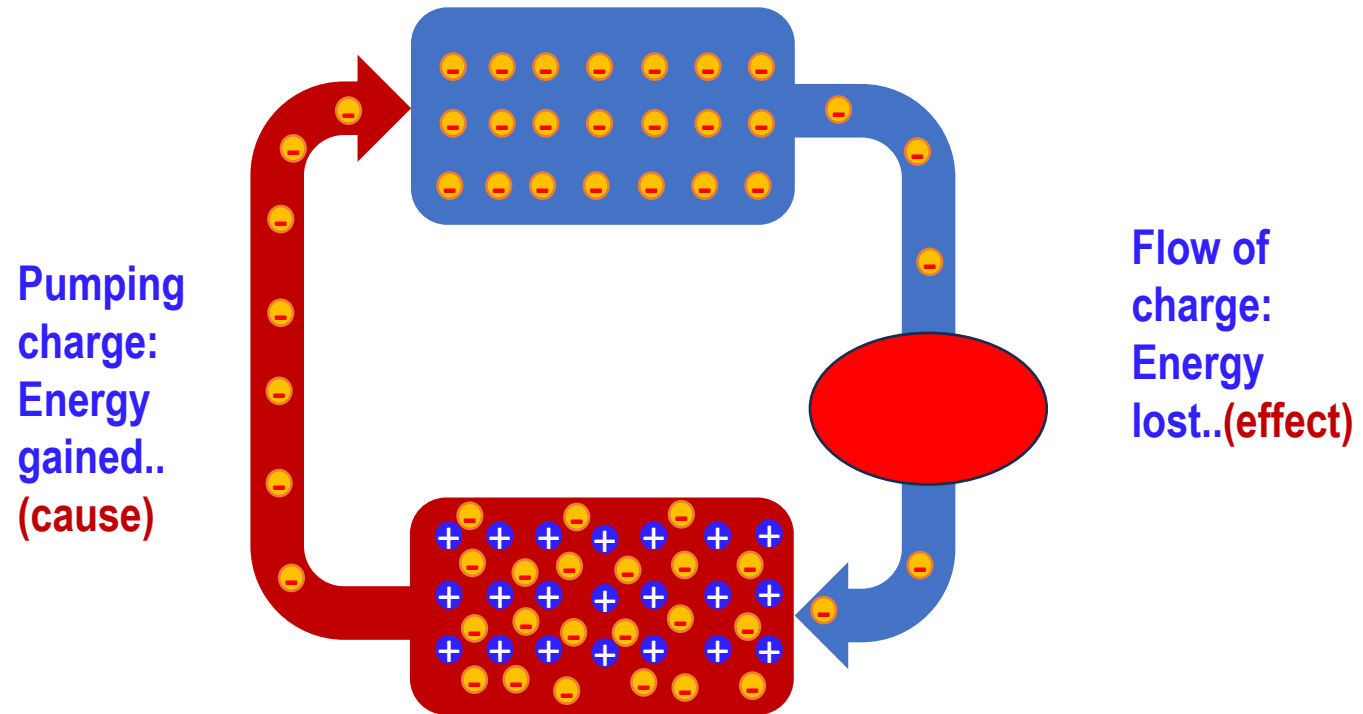
Situation three...continues..



- this is what the battery does, continuously pump charge to the negative piles using a chemical process thereby maintaining the potential difference between the two piles and at the same time maintaining the flow of charges.

So, finally we find closed network consisting of elements which generates some effect/response (flow of charge) as a result of certain cause (potential difference)

Situation three...continues..



two important ideas/concepts (measurable):

- the pressure that the electrons feel towards the pile of positive charges, and
- the transfer of electrons.

How do we define them?